

A PROJECT TITLE

ON

Soil Quality Classification Using Random Forest and Decision Tree

*project report submitted in partial fulfilment of the requirements for the
Course*

MACHINE LEARNING (25SC2107E)

BACHELOR OF TECHNOLOGY

in

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by

P. Naveen (2520030518)

J. Advait (2520030524)

P. Shashe Preetham (2520030534)

Under the Esteemed guidance of

Mr. G. Ravindra Babu

Professor, Department of CSE

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s 3 of the UGC Act, 1956)

Off Campus: Bachupally Gandamama Road, Bowrampet, Hyderabad, Telangana-500 043

Phone No: 7815625816, www.klef.edu.in



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Name of Title: Soil Quality Classification Using Random Forest and Decision Tree

1. Abstract:

Accurate assessment of soil quality is critical for sustainable agriculture, crop yield optimization, and land management planning. This project presents a machine learning-based approach to classify soil quality using two supervised tree-based algorithms: Decision Tree and Random Forest. The system follows a complete ML lifecycle, beginning with data collection and preprocessing of soil parameters such as pH, nitrogen, phosphorus, potassium content, organic matter, and moisture levels, followed by feature engineering, encoding, and missing value imputation to prepare the dataset for modeling.

A Decision Tree classifier is first constructed using Gini impurity and entropy-based splitting criteria, with pruning techniques applied to control overfitting and manage the bias-variance tradeoff. To improve predictive robustness, a Random Forest ensemble is then trained, leveraging bagging and feature subsampling to reduce variance and enhance generalization across unseen soil samples. Model performance is evaluated using key classification metrics including accuracy, precision, recall, F1-score, and the confusion matrix, with

cross-validation employed to ensure reliable results and prevent data leakage.

Feature importance analysis is conducted to identify the most influential soil attributes driving classification decisions, offering interpretable insights for agricultural stakeholders. The comparative study demonstrates that ensemble-based Random Forest outperforms a single Decision Tree in classification accuracy and stability, validating its suitability for real-world soil quality prediction systems.

2. INTRODUCTION:

Soil is the foundation of agricultural productivity, and its quality directly influences crop yield, resource utilization, and long-term land sustainability. Traditionally, soil quality assessment has relied on manual laboratory testing and expert interpretation, a process that is time-consuming, resource-intensive, and often inaccessible to small and marginal farmers. With soil health depending on multiple interacting parameters such as pH level, nitrogen, phosphorus, potassium, organic matter, and moisture, manual evaluation struggles to scale and often delays timely agricultural decisions.

This project proposes a machine learning-driven solution that automates soil quality classification using two tree-based supervised learning algorithms: Decision Tree and Random Forest. Unlike conventional lab-dependent methods, the proposed system delivers near-instant, explainable predictions once soil parameters are entered, making it scalable to large numbers of samples without added expert cost. The Decision Tree model offers an interpretable baseline built

On Gini impurity and entropy-based splitting, while the Random Forest model, an ensemble of multiple decision trees combined through bagging and feature subsampling, improves robustness and reduces the risk of overfitting.

By comparing both models using standard evaluation metrics such as accuracy, precision, recall, F1-score, and confusion matrices, this project identifies the most reliable classifier for real-world soil quality prediction. The resulting system supports informed decision-making for land and crop planning, smarter fertilizer and resource management, and improved agricultural productivity through data-driven insight.

3. SOFTWARE REQUIREMENTS:

- Programming Language: Python 3.10 or above
- IDE / Platform: Jupyter Notebook / Google Colab / VS Code
- Core Libraries: NumPy, Pandas (data handling and preprocessing)
- Machine Learning Library: Scikit-learn (Decision Tree, Random Forest, model evaluation)
- Visualization Libraries: Matplotlib, Seaborn (EDA and result visualization)
- Operating System: Windows 10/11, macOS, or Linux
- Dataset Format: CSV file containing soil parameters (pH, N, P, K, organic matter, moisture)


Signature of the Guide


Course Instructor